AI & COMPUTER-VISION BASED SURFACE & PRODUCT DEFECT INSPECTION

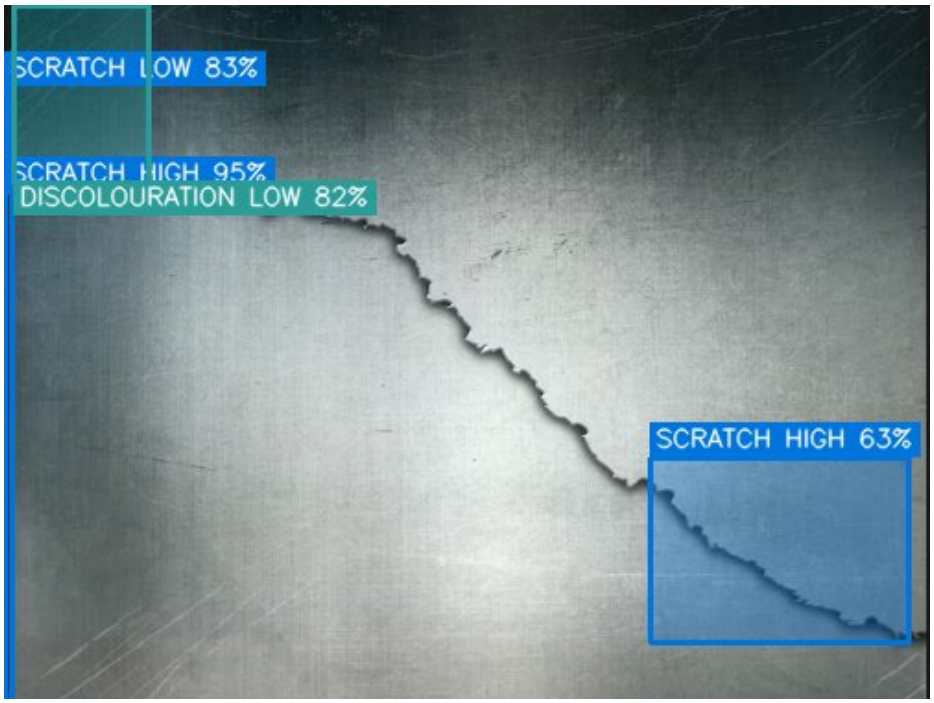
Automated Quality Inspection Report

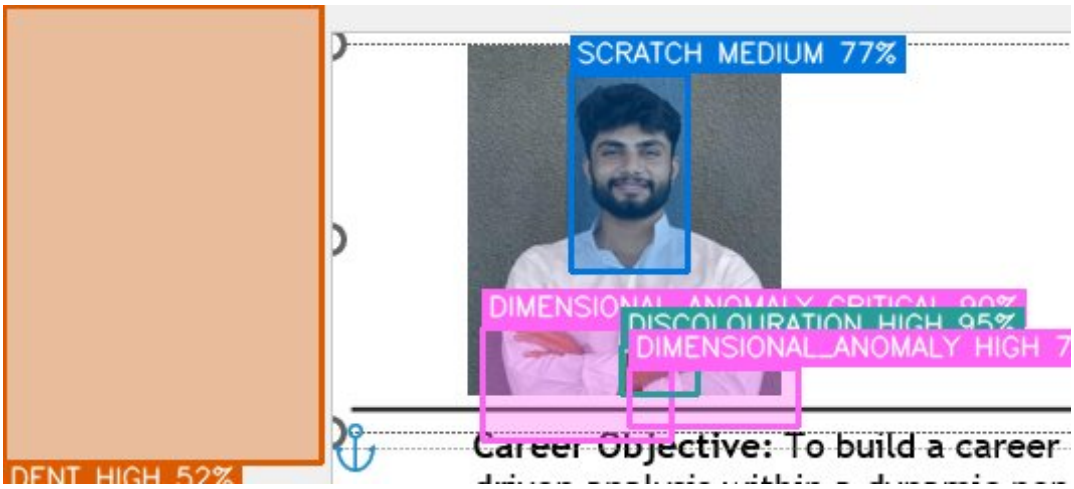
Inspection ID:	DEF-2026-0012	**Date & Time:**	2026-10-28 19:58:57
Product / Part:	Unnamed Product	**Batch / Lot:**	-
Product Line:	METAL SHEET	**Station:**	Line 1 · Final QC
Inspector:	Legal Metrology Inspector	**Outcome:**	**DEFECTIVE**

DEFECT SUMMARY

Defect Type	Count	Critical	High	Medium	Low
Scratch	4	0	2	1	1
Discolouration	2	0	1	0	1
Dimensional Anomaly	2	1	1	0	0
Dent	1	0	1	0	0

ANNOTATED VISUAL EVIDENCE





DETECTION LOG

#	Type	Severity	Confidence	Location (px)	Detector	Review
1	Scratch	HIGH	95%	0,101 - 4,376	OPENCV	PENDING
2	Scratch	HIGH	63%	350,245 - 490,344	ONNX	PENDING
3	Scratch	LOW	83%	0,44 - 3,94	OPENCV	PENDING
4	Discolouration	LOW	82%	5,0 - 78,96	OPENCV	PENDING
5	Dimensional Anomaly	CRITICAL	90%	238,160 - 332,216	OPENCV	PENDING
6	Discolouration	HIGH	95%	307,169 - 345,193	OPENCV	PENDING
7	Dimensional Anomaly	HIGH	74%	310,179 - 395,209	OPENCV	PENDING
8	Dent	HIGH	52%	0,0 - 158,227	ONNX	PENDING
9	Scratch	MEDIUM	77%	281,34 - 339,131	ONNX	PENDING

ROOT-CAUSE HINTS

Scratch (4)
• Abrasive contact during conveyor transfer or manual handling
• Burrs or debris on guide rails, fixtures, or tooling
• Insufficient protective film / interleaving between stacked parts
→ Inspect and deburr guide rails, grippers, and transfer chutes
→ Add protective film or separators at stacking and packing stations

Discolouration (2)
• Inconsistent curing, heat treatment, or coating thickness
• Oxidation / corrosion from moisture exposure or contamination
• Raw-material or paint batch variation
→ Check oven / curing temperature uniformity and dwell time
→ Review storage humidity and batch traceability of coating material

Dimensional Anomaly (2)
• Tool or die wear, mould shrinkage, or fixture misalignment
• Machine calibration drift or incorrect program offsets
→ Run tool-wear inspection and re-calibrate the machine against the master gauge
→ Increase in-process gauging frequency until the deviation is resolved

Dent (1)
• Impact during handling, drops at transfer points, or improper stacking
• Foreign particles trapped between die and workpiece
→ Add cushioning at drop points and check palletisation load limits
→ Clean dies / moulds and add air-blow before forming

DETERMINATION & SIGN-OFF

Inspector Remarks: Defect inspection reviewed and signed off by quality inspector

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Analysed by: Defect Inspection Engine (ONNX+OPENCV)

Quality Sign-off:
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